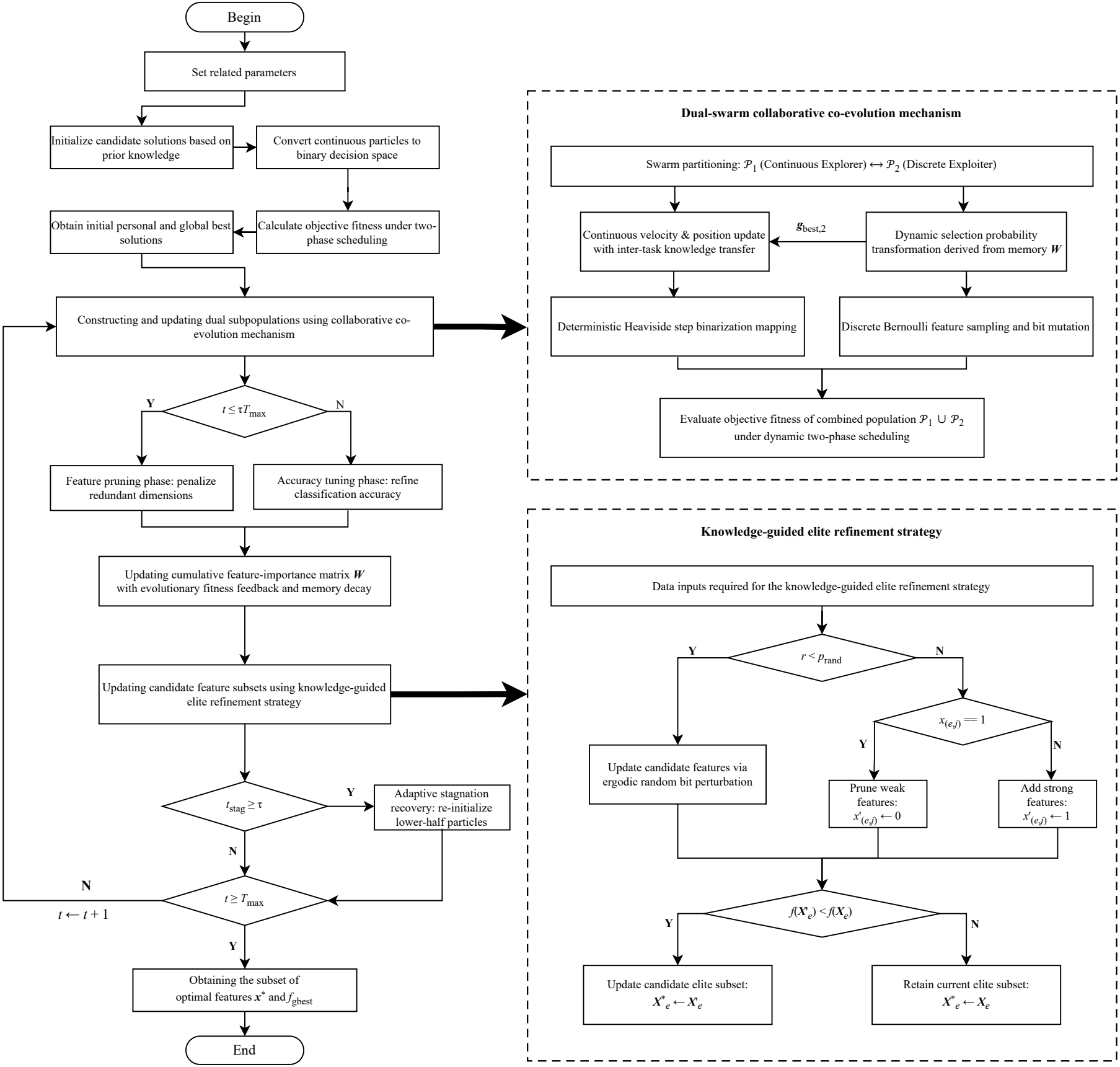

Begin
Set related parameters
Initialize candidate solutions based on prior knowledge
Convert continuous particles to binary decision space
Obtain initial personal and global best solutions
Calculate objective fitness under two-phase scheduling
Constructing and updating dual subpopulations using collaborative co-evolution mechanism
Y
$t \le \tau T_{\max}$
N
Feature pruning phase: penalize redundant dimensions
Accuracy tuning phase: refine classification accuracy
Updating cumulative feature-importance matrix $\boldsymbol{W}$ with evolutionary fitness feedback and memory decay
Updating candidate feature subsets using knowledge-guided elite refinement strategy
$t_{\text{stag}} \ge \tau$
Y
Adaptive stagnation recovery: re-initialize lower-half particles
N
$t \ge T_{\max}$
N
$t \leftarrow t + 1$
Y
Obtaining the subset of optimal features $\boldsymbol{x}^*$ and f_{gbest}
End
Dual-swarm collaborative co-evolution mechanism
Swarm partitioning: $\mathcal{P}_1$ (Continuous Explorer) ↔ $\mathcal{P}_2$ (Discrete Exploiter)
Continuous velocity & position update with inter-task knowledge transfer
$\boldsymbol{g}_{\text{best},2}$
Dynamic selection probability transformation derived from memory $\boldsymbol{W}$
Deterministic Heaviside step binarization mapping
Discrete Bernoulli feature sampling and bit mutation
Evaluate objective fitness of combined population $\mathcal{P}_1 \cup \mathcal{P}_2$ under dynamic two-phase scheduling
Knowledge-guided elite refinement strategy
Data inputs required for the knowledge-guided elite refinement strategy
Y
$r < p_{\text{rand}}$
N
Update candidate features via ergodic random bit perturbation
$x_{(e,j)} == 1$
Y
N
Prune weak features: $x'_{(e,j)} \leftarrow 0$
Add strong features: $x'_{(e,j)} \leftarrow 1$
Y
$f(\boldsymbol{X}'_e) < f(\boldsymbol{X}_e)$
N
Update candidate elite subset: $\boldsymbol{X}^*_e \leftarrow \boldsymbol{X}_e$
Retain current elite subset: $\boldsymbol{X}^*_e \leftarrow \boldsymbol{X}_e$